PulseBand: Design Prototype Documentation

Team 16: PulseBand

Andhika Nayaka Arya Wibowo (s49847008)

Serafina Nala Putri Setiawan (s4980036)

Nguyen Minh Thao Vy (s47289006)

Syifa Ananda Widyati (s49800272)

The University of Queensland

DECO3500: Social & Mobile Computing

GitHub repository

https://github.com/terserahdehh/DesignProposal---DECO3500

Design Process Overview

PulseBand · Team 16 · DECO3500

Explore the outcome: A2 poster · Brochure and videos · Promotional website.

Try the prototype: Wearable simulator · Physical-band companion · Setup and deployment. No account or login is required. For a physical demonstration, use the configured Nayaka–Syifa pair and quick-start instructions.

Opportunity and research

PulseBand explores how friends can communicate changing plans and reconnect at concerts without repeatedly checking phones. We asked how social and mobile technology might support connection while respecting attention and the freedom to separate. Our outcome combines prepared wrist messages, approximate friend-finding and a deliberate peer SOS alert.

We combined a 27-response survey, five interviews, online observations and secondary research. Finding friends was the greatest difficulty for 13 respondents (48.1%), followed by communication for eight (29.6%). Ten preferred approximate direction or distance, and five wanted quick updates. These descriptive preferences prioritised features, but do not establish population demand or prototype effectiveness (Team 16, 2026d).

Interviews connected poor reception, inaudible calls, moving crowds and group coordination. Participants used landmarks and meeting points as workarounds. One reported that a location message could become obsolete before arriving (Team 16, 2026c, pp. 2–3). Communicating intentions under changing conditions became central. Convenience sampling, retrospective accounts and online examples limit generalisation.

Lindström and Pettersson (2010) identify ad-hoc re-planning and shareability as festival design challenges. This explains why a message should communicate an intended action, beyond a status. Larsen et al. (2013) used fixed Bluetooth scanners to study festival

movement and grouping, while acknowledging discoverability bias. This contextualises reconnection, but does not establish PulseBand's radio range or positioning accuracy.

Translating evidence into interaction

We narrowed the concept to Find Friend, quick messages and SOS. Prepared messages address inconvenient typing and loud environments. Names and text accompany colour so meaning does not depend on memorising lights. Message composition moved to a larger companion screen, leaving brief selection and confirmation on the wrist. This trades preparation effort for reduced interaction during the performance. Earlier explorations of lyrics, memory bookmarking and vibration did not become physical features.

Ellis et al.'s (1991) groupware perspective distinguishes communication, coordination and collaboration. Delivery communicates information, deciding who waits coordinates action, and reuniting depends on cooperation. Asynchronous updates may be read later, so acknowledgement means device receipt, not attention or agreement. An arrow supplies awareness but cannot negotiate a meeting plan.

Dagan et al.'s (2019) social-wearables framework asks who triggers an input, how intentional it is, and who perceives the output. PulseBand uses deliberate touch for messages and opt-in location updates. Visual outputs support brief awareness but may also be visible to bystanders. The framework therefore prompts attention to control, consent and display visibility. It focuses on co-located interaction. We apply its questions to temporary separation within a shared venue, without claiming equivalent evaluation results.

Figure 1

Early concept artefact showing the breadth of exploration before the physical prototype focused on communication and reconnection.

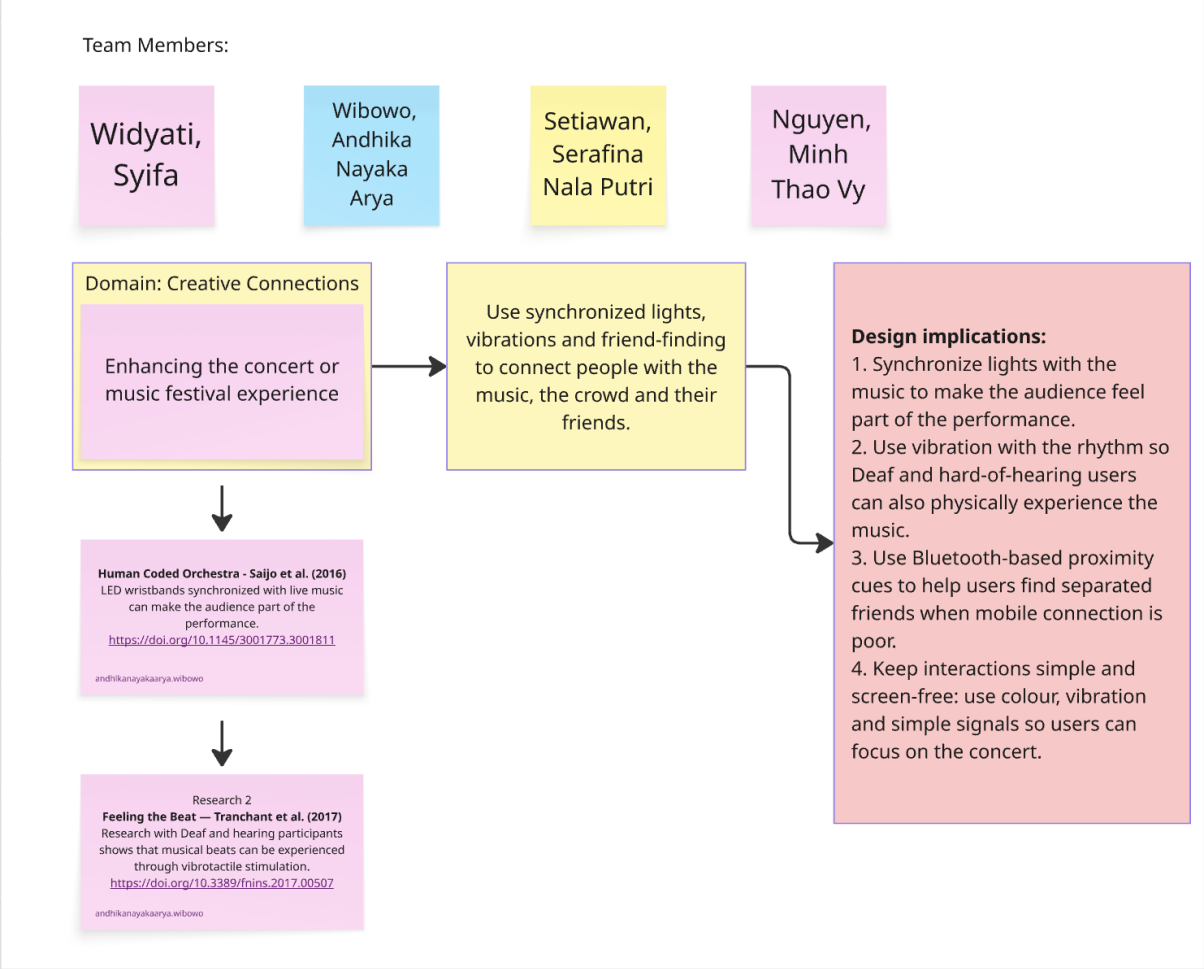


Figure 2

Team-supplied framework image reproducing the input/intentionality and input/output matrices from Dagan et al. (2019, p. 1005, Figures 1–2). Original figure and icon credits remain visible. Copyright 2019 by the authors, with publication rights licensed to ACM. These are prior-work matrices, not PulseBand evaluation results.

Design Framework for Social Wearables

https://dl.acm.org/doi/pdf/10.1145/3322276.3322291

https://dl.acm.org/doi/10.1145/3322276.33222...
ACM Digital Library | Updated 03-09-2026 @ 10:04 GM...

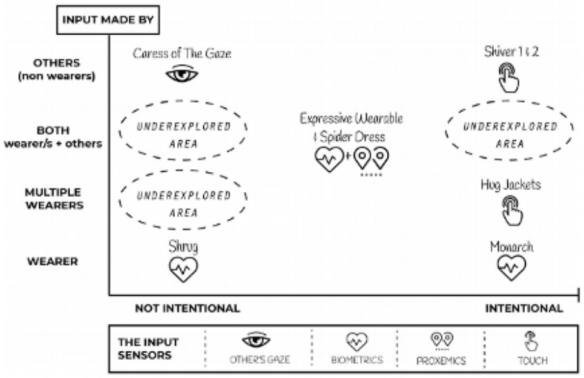


Figure 1. Matrix of examples depicting who is responsible for triggering the input, the degree of their intentionality, the input sensors, and the underexplored areas. Icons credit: Noun Project {icongeek; iconsphere; Marek Polakovic; Ramesha}.

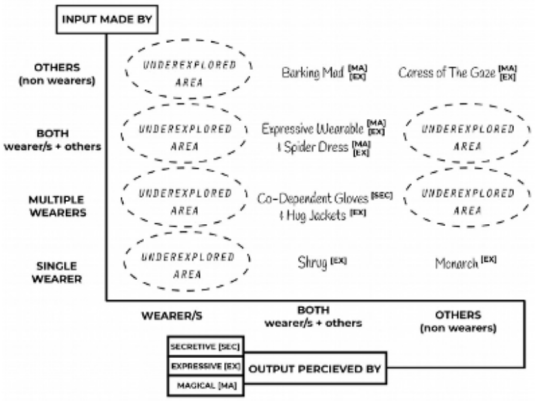


Figure 2. Matrix visualizing who "is sensed" and who "senses" the wearable, (i.e. can perceive its output), the kind of output (expressive, secretive or magical), and underexplored areas.

Prototyping and early evaluation

The early browser prototype simulated messages, location and group connection. Nayaka, Amy, Serafina and Syifa collected concept feedback through demonstrations and scenario walkthroughs (Team 16, 2026a). Participants valued prepared messages and customisation but needed explanations of controls, colours and recipients. "Taking a break" was ambiguous: it could invite company or request privacy. Others wanted permission before being found and warned that exact-looking distances implied excessive accuracy.

These findings supported a short menu, identifiable senders and recipients, editable messages and explicit delivery states. Three separate OK presses make SOS deliberate, with cancellation before sending. However, this adds effort under stress and has not been validated as an accessible emergency interaction. Dedicated replies and permission for each Find Friend request remain unimplemented. Turchet et al. (2021) informed our interest in personalised tactile feedback. Their different musical-haptics context does not validate our communication design, and our boards have no vibration motor.

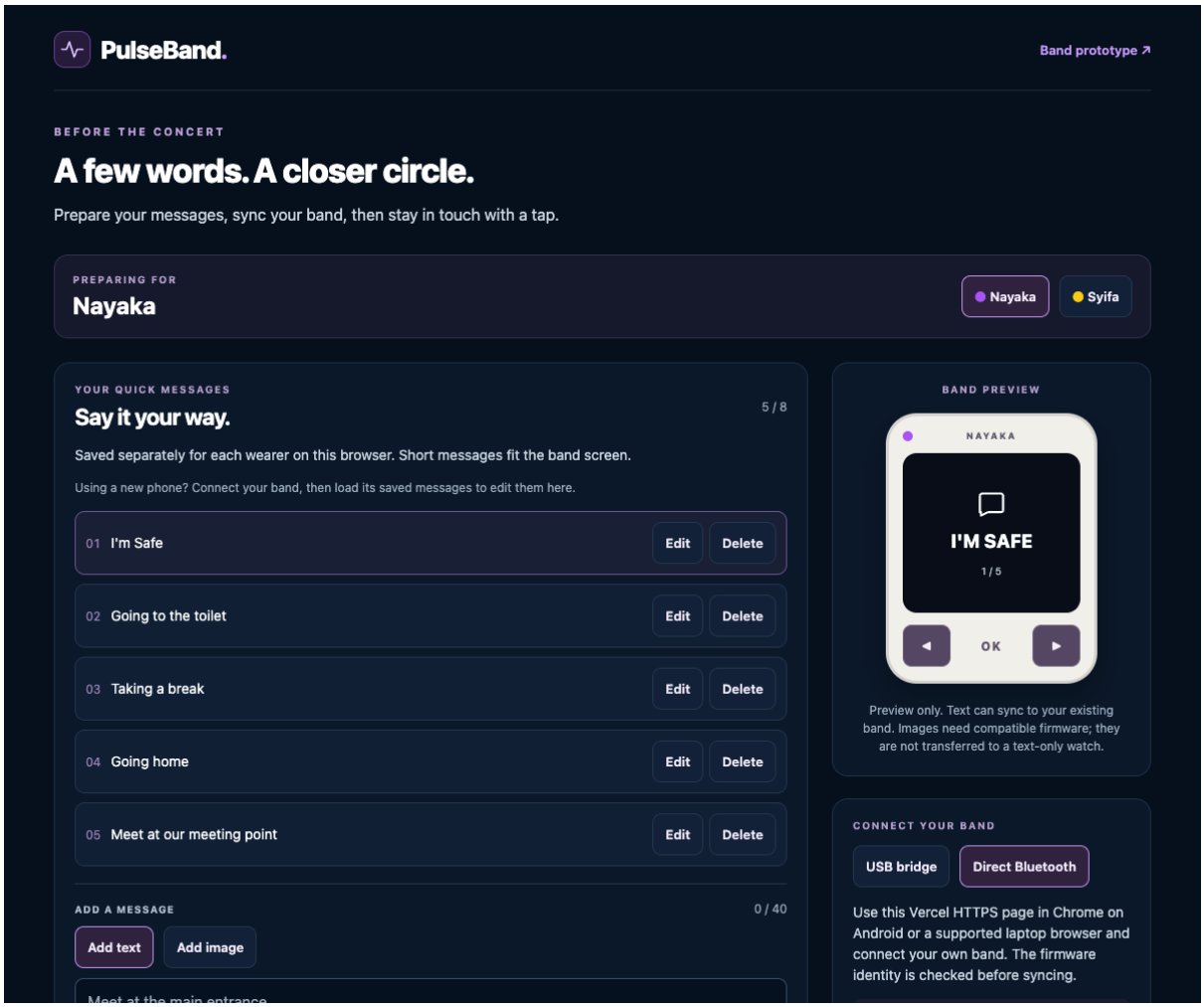
From simulation to a physical pair

Our retrospective Week 9 record reports tutor concern that friend-finding remained simulated. We prioritised two ESP32 touch-display bands. Bluetooth connects the companion for configuration, while ESP-NOW exchanges messages between the Nayaka–Syifa pair. Prepared text persists on each band. Larger groups remain simulated. The companion stores image presets, but the documented installed watches remain text-only.

The boards have neither GPS nor a compass. Independent phone locations supply approximate distance and a north-referenced bearing that does not rotate with the wrist. Stale or insufficiently separated fixes suppress direction. This exposes uncertainty, but provides no indoor positioning, safe route or metre-level movement tracking. Messaging avoids mobile data dependence, while guidance still depends on phone location quality.

Figure 3

Companion interface captured from the production website on 25 October 2026, before connecting hardware. This shows message preparation and transport selection. It is not a live hardware or location test.



Physical evaluation and learning

Five participants subsequently tried Find Friend using two physical bands (Team 16, 2026b). Their discussion identified useful direction cues, inconsistent guidance during sideways movement, subtle brightness changes and unclear proximity dots. The transcript supports these qualitative findings, but records no controlled task times, success rates or accuracy measurements. It cannot establish festival reliability. Participant labels belong to this session and are not assumed to match the interview sample.

We therefore prioritised clearer directional meaning, stronger proximity feedback and fewer competing indicators. Faster blinking and removing dots are recorded as proposed refinements, not universally verified implementation. Engineering safeguards address stale locations and acknowledgement errors, while automated browser tests check interface and protocol behaviour. Neither substitutes for user evaluation. The next study should compare cue interpretations with independent location sources, observe safe movement, and assess visibility, comfort, refusal and message ambiguity with more diverse attendees.

Exhibition

The supplied photographs document the exhibit display: poster, brochures, promotional website, companion interface and two physical boards. This brought the research narrative and interactive artefacts together for visitors. The group photograph adds visual context to the team-reported Week 12 participation. These images document the exhibit, without supplying task-success measurements or a record of the oral pitch.

Figure 4

Team-supplied screenshot of the exhibition setup, showing the poster, brochures, promotional website, companion interface and two boards.



Figure 5

Team-supplied exhibition group photograph showing the display and people gathered around the demonstration table.



Contributions and critical reflection

All four members contributed research and synthesis. Amy (Nguyen Minh Thao Vy) conducted Interviews 1–2 and scenario evaluation, created the brochure and video, and revised content and poster. Serafina conducted Interview 3 and early testing, co-created the website and poster, edited video and assisted coding/hardware. Syifa conducted Interview 5 and an early walkthrough, co-created the website/poster and assisted coding/hardware. Nayaka conducted Interview 4 and early testing, and led coding, hardware integration and deployment. Coordination used WhatsApp and class discussions.

Technical connection and social coordination are different achievements. A credible demonstration exposes uncertainty and leaves room to decline. Ethical Considerations applies this to privacy, accessibility, peer alerts and publication. Our documented cycle follows the course brief (The University of Queensland, 2026), with remaining limitations and historical evidence gaps stated explicitly.

Ethical Considerations

PulseBand · Team 16 · DECO3500

Purpose and framework

We want PulseBand to help friends stay connected without making anyone feel they have to be tracked. The design also needs to be clear about its limits: an approximate arrow is not a safe walking route, and an SOS to a friend does not contact emergency services.

For this analysis, we use the **Ethical Disclaimer** tool (Ethics for Designers, n.d.). Its five steps consider the current situation, stakeholders, intentions, unethical situations and the responsibilities designers accept. The sections below apply those steps to our own prototype, rather than treating a disclaimer as permission to transfer harm to users.

Loke and Matthews' (2020) *Scaffolding of Interaction Design Education Towards Ethical Design Thinking* treats ethics and values as part of interaction design education. Its published abstract supports integrating value-sensitive reflection into design. We use the Ethical Disclaimer to structure our analysis of stakeholders, risks and design responsibilities.

1. Current situation

Friends at loud, crowded events coordinate using gestures, group chats, landmarks and agreed checkpoints. Interviewees describe delayed messages, difficulty hearing calls and uneven coordination responsibilities. Separation may be intentional and enjoyable. Constant physical togetherness is not the design goal. The wearable is optional support for occasional communication and reconnection.

The implemented hardware is one fixed two-band pair. Phone location is required for guidance. The boards have no GPS, magnetometer or vibration motor. Location can be inaccurate or suspended by the browser. Messages need powered bands within ESP-NOW range. These conditions affect what a wearer may reasonably trust.

2. Who could be affected

Table 1

Stakeholders, benefits and potential harms.

Stakeholder	Benefit or interest	Potential burden or harm
Wearer sending messages or location	Fast communication and reconnection	Loss of privacy, setup burden, misleading delivery or guidance
Friend receiving a signal	Awareness and ability to coordinate	Ambiguous intentions, pressure to respond, anxiety from false alerts
Attendees with different sensory/motor needs	Alternative visual/tactile channels	Unreadable colours, small controls, uncomfortable vibration or inaccessible gestures
Other attendees and bystanders	Less phone obstruction	Collisions or disruption from screen-focused movement
Venue staff and emergency responders	Clear escalation expectations	Assumed responsibility for an alert they never receive
Research participants	Influence the design voluntarily	Exposure of identity, recordings or sensitive disclosures
Team, teaching staff and future maintainers	Understandable evidence and implementation	Unsupported claims, exposed keys or incomplete evidence
Venue operators and the environment	Reusable optional devices	Cost, maintenance, battery handling and electronic waste

3. What we want the design to support

Our aim is to keep messages quick, make location uncertainty visible and let people choose when to share. Typing happens on the companion page so the wrist interaction can stay short. People should be able to take a break or do something separately without feeling they are letting the group down.

Our prototype should let visitors distinguish demonstration effects from real communication. Device acknowledgement must not be described as a wearer reading or

agreeing with a message. Neither simulated nor physical SOS should imply staff or emergency-service integration.

4. Risks and design responses

Location privacy. Finding friends should not turn into an expectation that everyone stays trackable. Users can start and stop sharing through the companion page. The watch does not yet let someone accept or refuse each finding request.

Dagan et al. (2019) ask who activates a wearable and who perceives its output. Applied here, intentional permission to start sharing does not authorise every later finding request, and a wrist display may still be visible to bystanders. Lindström and Pettersson (2010) also identify control over shared group information as a design concern. Our future group controls should therefore support selective sharing and refusal, rather than assume friendship establishes permanent permission.

Misleading direction. An arrow can look trustworthy even when the phone location is inaccurate. The band hides direction when the location is too old or uncertain, but it cannot show a safe route around barriers or through a crowd.

Misunderstood messages. A message reaching the band does not mean the person has read it or agreed to act. Friends may still need to confirm whether they will wait, move or meet somewhere else.

SOS expectations. Three OK presses help avoid accidental alerts, but could be difficult when someone is stressed. SOS only alerts the paired friend. It does not contact venue staff or emergency services.

Accessibility and attention. Small controls, colours and screen brightness may work differently for different users. The interface includes text alongside colour, but the physical prototype has no vibration motor. People also need to stay aware of the crowd rather than keep watching the display while walking.

Personal data. Research documents, location information and connection details can reveal more about someone than intended. Public documents need careful handling, and private credentials do not belong in the repository.

Comfort and reuse. A wearable needs to be comfortable and affordable enough for people to use. Loaning or reusing devices could reduce cost and waste, but these arrangements are outside our prototype.

5. Responsibilities and boundaries

Our responsibility as a team is to explain what works, make errors understandable and give users control over sharing and cancellation. We also need to describe our tests accurately and protect participant information. The technical limits matter because people may act on what the screen tells them.

Shared responsibilities include agreeing a checkpoint and interpreting a message together. Users should be able to decide whether to move, wait or decline. Venue staff retain their existing roles. A peer alert does not delegate a response to them. We should design clear expectations rather than shift every burden to a main group organiser.

Some conditions are outside our control, including phone location accuracy, radio interference, the browser pausing updates and barriers at the venue. The interface needs to make problems visible so friends can fall back on a meeting point or another way to contact each other.

Location and stored data

Location sharing starts when the user chooses it on the companion page. Each wearer needs to share their own phone's location. The band stops using a location after 60 seconds without an update.

Stopping sharing ends new updates and asks the band to clear its location. If the connection fails, the previous location may remain until it expires. Location is not saved permanently on the band. Prepared messages are saved so they remain available after restarting it.

The web prototype also remembers the selected identity and message lists in the browser. These names are demonstration labels, not secure accounts. A finished product would need stronger account and group privacy controls.

Larsen et al. (2013) inferred festival movement and grouping from time-stamped Bluetooth observations. This demonstrates the analytical value of repeated location-related records, while also motivating our decision to avoid retaining a location history. Their study's historical consent arrangements do not establish permission for our own collection or publication. Our ethics assessment concerns what PulseBand users choose to share and what our team publishes.

Research and public documentation

The source interviews include consent to recording and university-project use, but that does not by itself establish consent to identifiable public publication. This wiki uses participant codes and summarises findings. The combined interview PDF and retained evaluation document contain participant details, so publication raises privacy and consent considerations beyond participation in the study. Recordings, precise test locations, private credentials and raw Bluetooth/session logs should not be included in public artefacts.

For further evaluation, explain purpose, procedures, recording, data access and withdrawal before obtaining consent. Use a safe setting, comfortable sound levels and voluntary sharing. A participant should be able to decline location or SOS tasks without penalty. Store raw notes within the agreed access scope. Publish de-identified results and clearly state limitations. Avoid portraying one person's sensory preferences as representative of a whole community.

Priorities and evidence needed

The following priorities are our design judgement, not measured risk probabilities. Existing controls reduce particular risks. No row implies the remaining risk has been eliminated.

Table 2

Ethical priorities, current controls and remaining responsibilities.

Risk and priority	**Current response**	**Remaining responsibility and acceptance evidence**
Coercive tracking (high)	Sharing starts and stops on the companion. Fixes expire.	Per-request refusal on the watch is absent. Team must evaluate whether a wearer can decline without social pressure. Do not describe browser permission as ongoing social consent.
Misleading guidance (high)	Stale/uncertain fixes suppress the arrow. North reference is explained.	Team must test whether unfamiliar users recognise uncertainty and avoid treating an arrow as a safe route. Retain agreed meeting points as a fallback.
False confidence in SOS (high)	Three confirmations, cancellation before sending, peer-only destination.	Test confirmation effort and recipient interpretation under safe simulated stress. Do not advertise an emergency response guarantee.
Public research exposure (high)	Wiki summaries use participant labels.	Team must review identifiable source documents and faces/voices in videos, and verify permission for public publication. Recording consent alone is insufficient. Public Git history may retain earlier versions.
Exclusion through colour or touch (medium)	Names and text supplement colour. Menu remains short.	Observe low-vision, colour-vision and motor-access needs with consenting participants. Physical vibration is absent, so do not claim an evaluated haptic or universally accessible experience.
Notification burden and ambiguous intentions (medium)	Editable messages and sender identity provide context.	Test mutually understood messages and the ability to ignore or decline. Dedicated replies remain future work.
Comfort, cost and waste (medium)	Prototype can be reused for demonstrations.	Team must evaluate fit, safe handling, cleaning, maintenance and eventual disposal before wider deployment. No lifecycle or affordability study has been completed.

Evaluation and accountability

Before a further study, explain which functions are real and which are simulated, who can access recordings, whether any material will be public, and how a participant can stop or withdraw within the agreed study procedure. Separate participation, recording and public-release choices. Do not require location sharing to participate in interface testing. A stationary, supervised task can assess interpretation without asking people to follow uncertain cues through a crowd.

For each task, record the consent conditions, observed misunderstanding, control used and unresolved consequence. Useful acceptance evidence would show a wearer stopping sharing, both friends interpreting an acknowledgement correctly, and a visitor identifying SOS as a peer alert. These are proposed criteria for the next evaluation, not outcomes already demonstrated with participants. All four team members share responsibility for reviewing public evidence and explaining limits at the exhibit. Coding ownership does not transfer that obligation to one person.

This assessment is a living design document. The move from simulation to phone-derived location made privacy, uncertainty and device acknowledgement more concrete. Adding larger groups, account identity, venue integration or stronger location technology would require another stakeholder and data-flow review. Loke and Matthews (2020) provide the broader educational argument for making values explicit throughout design. Our particular controls and priorities are our application to PulseBand.

References

Dagan, E., Márquez Segura, E., Altarriba Bertran, F., Flores, M., Mitchell, R., & Isbister, K. (2019). Design framework for social wearables. In *Proceedings of the 2019 on Designing Interactive Systems Conference* (pp. 1001–1015). Association for Computing Machinery. https://doi.org/10.1145/3322276.3322291

Ellis, C. A., Gibbs, S. J., & Rein, G. (1991). Groupware: Some issues and experiences. *Communications of the ACM, 34*(1), 39–58. https://doi.org/10.1145/99977.99987

Ethics for Designers. (n.d.). *Ethical disclaimer*. https://www.ethicsfordesigners.com/ethical-disclaimer

Larsen, J. E., Sapiezynski, P., Stopczynski, A., Mørup, M., & Theodorsen, R. (2013). *Crowds, Bluetooth, and rock-n-roll: Understanding music festival participant behavior* [Preprint]. arXiv. https://doi.org/10.48550/arXiv.1306.3133

Lindström, S., & Pettersson, M. (2010). Supporting ad-hoc re-planning and shareability at large-scale events. In *Proceedings of the 16th ACM International Conference on Supporting Group Work* (pp. 245–252). Association for Computing Machinery. https://doi.org/10.1145/1880071.1880113

Loke, L., & Matthews, B. (2020). Scaffolding of interaction design education towards ethical design thinking. In G. Melles (Ed.), *Design thinking in higher education: Interdisciplinary encounters* (pp. 165–181). Springer. https://doi.org/10.1007/978-981-15-5780-4_8

Team 16. (2026a). *Early prototype evaluations* [Research index]. https://github.com/terserahdehh/DesignProposal---DECO3500/blob/main/research-findings/README.md

Team 16. (2026b). *Friend-finding usability testing* [Transcript and summary]. https://github.com/terserahdehh/DesignProposal---DECO3500/blob/main/research-findings/team/usability-testing-summary.md

Team 16. (2026c). *Interview transcripts* [Five research interviews]. https://github.com/terserahdehh/DesignProposal---DECO3500/blob/main/research-findings/interview-transcripts.pdf

Team 16. (2026d). *Survey results* [Aggregate data and synthesis]. https://github.com/terserahdehh/DesignProposal---DECO3500/blob/main/research-findings/team/README.md#survey-results

The University of Queensland. (2026). *Assignment 3: Design project (sequence)* [DECO3500 Social & Mobile Computing course assignment brief, Semester 2].

Turchet, L., West, T., & Wanderley, M. M. (2021). Touching the audience: Musical haptic wearables for augmented and participatory live music performances. *Personal and Ubiquitous Computing, 25*(4), 749–769. https://doi.org/10.1007/s00779-020-01395-2